# 5 Tips Optimize Python Pandas

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1. Vectorize Operations
2. Groupby Efficiently
3. Use Inner Join
4. Filter First then Join
5. Sort then Merge

df

| C1 | C2 | C3 |
|---|---|---|
| 1 | 3 | 1.5 |
| 2 | 3 | 1.5 |
| 3 | 3 | 2 |
| 4 | 3 | 1.5 |



new_df

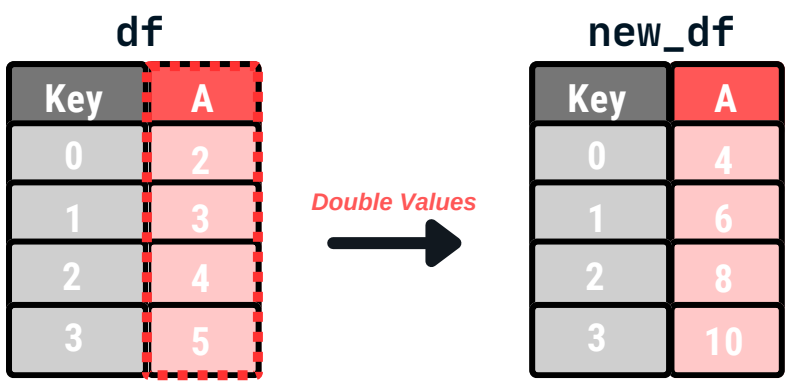
| C1 | C2 | C3 |
|---|---|---|
| 1 | 3 | 1.5 |
| 2 | 3 | 1.5 |
| 3 | 3 | 2 |

# Tip 1

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# *Vectorize Operations*

🎯 **Task** - Multiply each integer value by 2 in column “A”




❌ **Inefficient Solution** - Use loops to iterate across rows

```
for idx in range(len(df)):
    df.loc[idx,'A'] *= 2
```

✅ **Efficient Solution** - Use vectorized methods

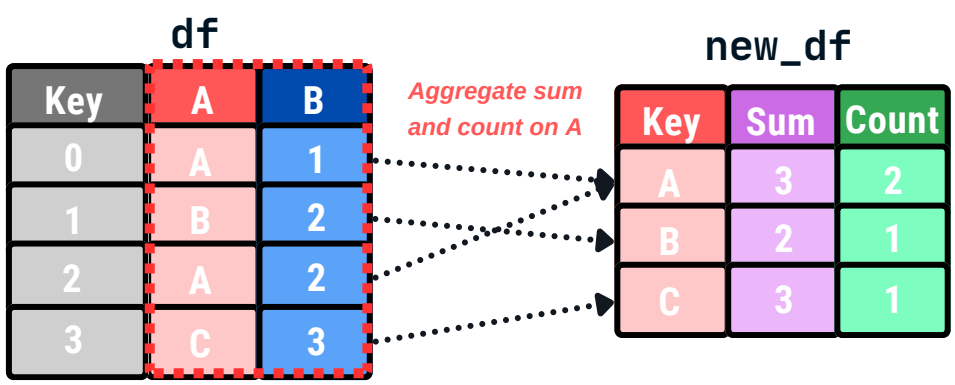
```
df['A'] = df['A'] * 2
```

# Tip 2

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# *Use Groupby Efficiently*

🎯 **Task** - Aggregate sum and count on column B given column A




❌ **Inefficient Solution** - Apply **groupby** separately

```
sums = df.groupby('A')['B'].sum()
count = df.groupby('A')['B'].count()
```

✅ **Efficient Solution** - Combine **groupby** using **agg()**

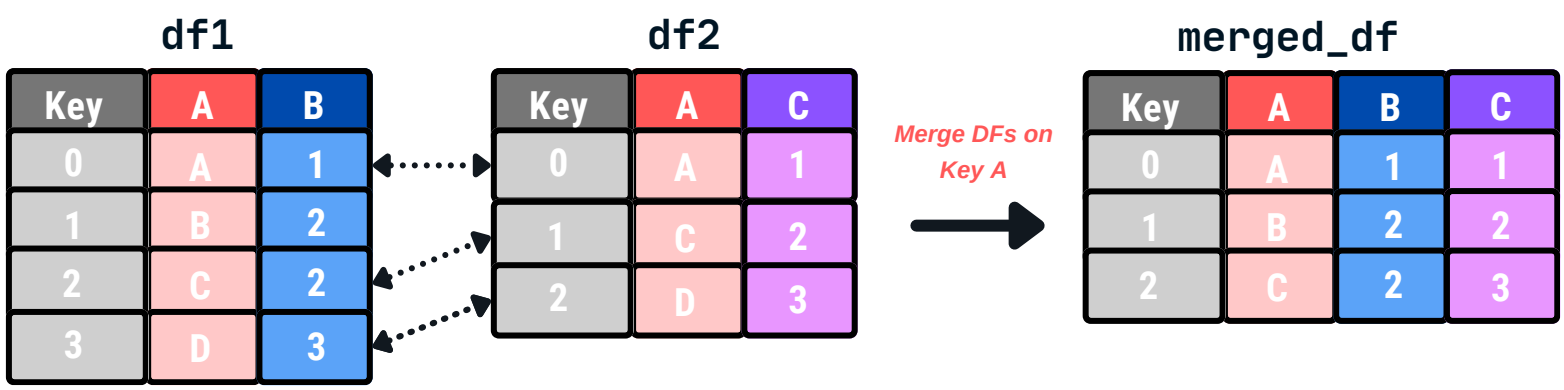
```
df.groupby('A')
  .agg({'C': [sum, count]})
```

# Tip 3

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# *Use Inner Join*

🎯 **Task** - Merge df1 and df2 on common values of A




❌ **Inefficient Solution** - Outer merge then filter

```
merged_df = df1.merge(df2, on='A', how='outer')
merged_df = merged_df[merged_df['A'].notnull()]
```

✅ **Efficient Solution** - Inner merge to retain common values

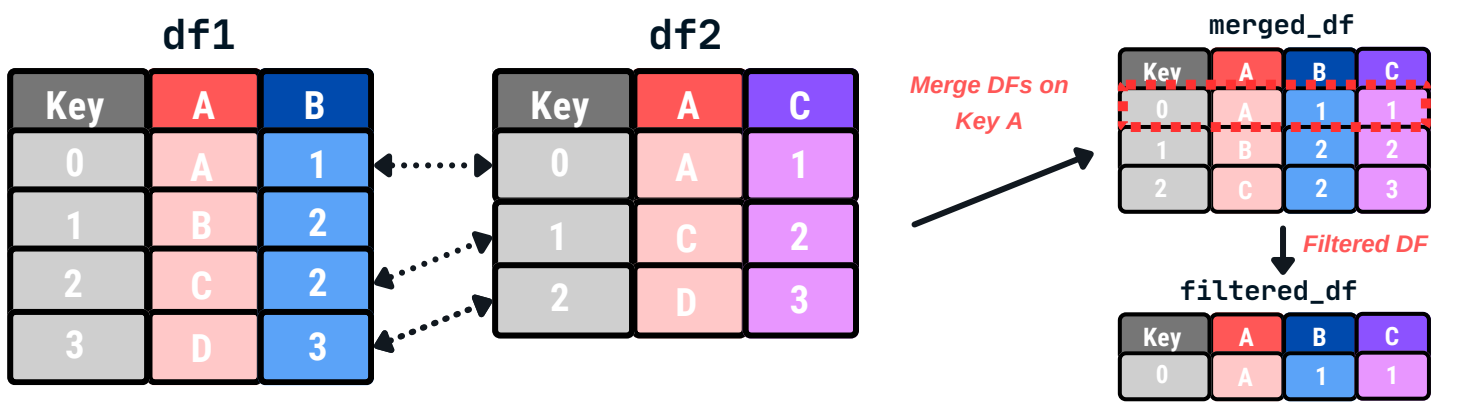
```
merged_df = df1.merge(df2, on='A', how='inner')
```

# Tip 4

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# *Filter First Then Join*

🎯 **Task** - Select C = 1 on joined tables of DF1 and DF2




❌ **Inefficient Solution** - Merge then filter

```
merged_df = df1.merge(df2, on='A', how='inner')
filtered_df = merged_df[merged_df['C'] = 1]
```

✅ **Efficient Solution** - Filter for a smaller DF, then merge

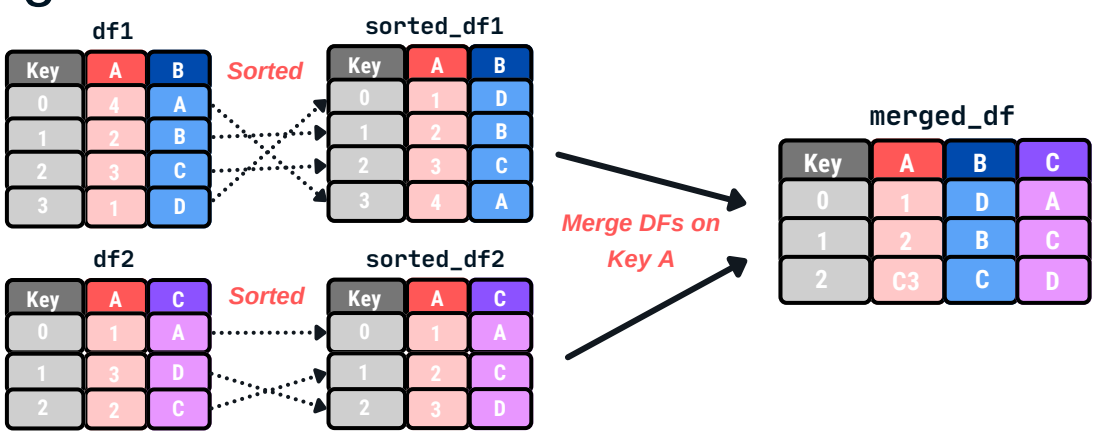
```
filtered_df = df2[df2['C'] = 1]
merged_df = df1.merge(filtered_df, on='A', how='inner')
```

# Tip 5

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# *Sort then Merge*

## 🎯 **Task** - Merge DataFrames on column A




## ❌ **Inefficient Solution** - Merge

```
merged_df = df1.merge(df2, on='A', how='inner')
```

## ✅ **Efficient Solution** - Filter for a smaller DF, then merge

```
sorted_df1 = df1.sort_values('A', ascending=True)
sorted_df2 = df2.sort_values('A', ascending=True)
merged_df = sorted_df2.merge(sorted_df1, on='A', how='inner')
```